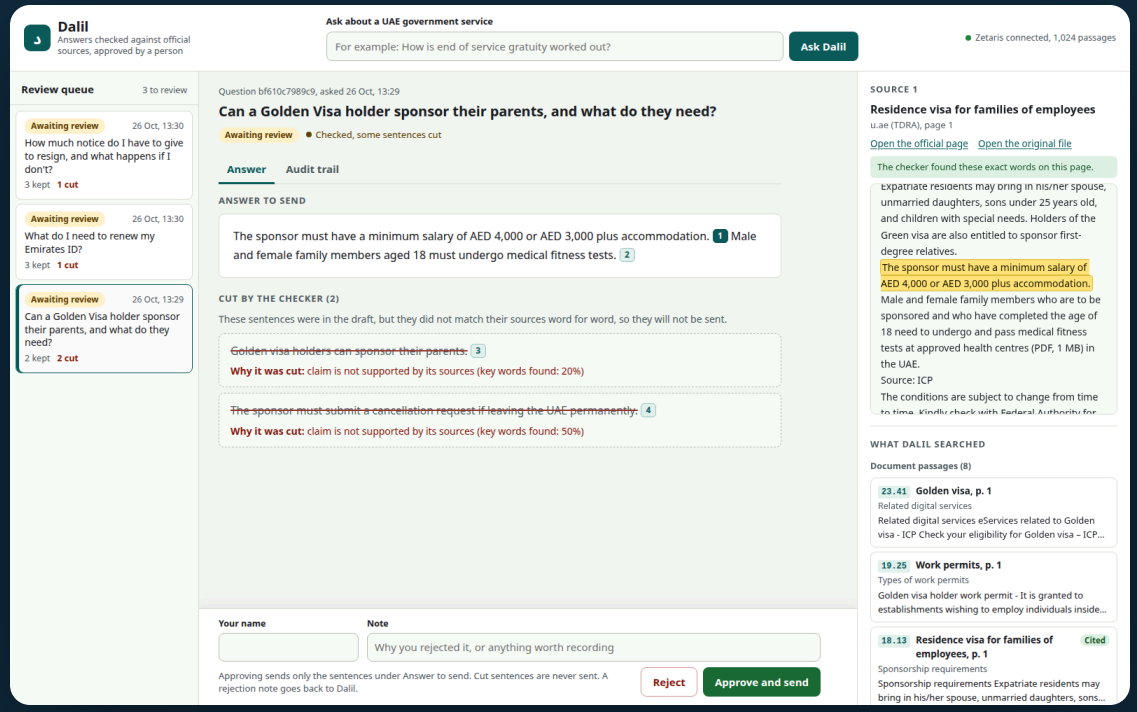
Dalil

Answers about UAE government services that you can check.



Team LFG: Kaif Ahmed, Abdul Samad, Abdul Sami Khawaja

Track 02, The Agent That Can Explain Why

Open Agent Hackathon 2026

PROBLEM AND USERS

One wrong answer costs a resident a fine

- Residents ask the same questions every day: sponsoring family, renewing an Emirates ID, notice periods, end-of-service gratuity.
- The answers sit across service pages, federal laws and fee tables from different authorities.
- A wrong answer means a fine or a rejected application, so a service team cannot let a chatbot answer unchecked.
- **Who uses Dalil:** service-centre staff who answer residents, and the residents who get the reviewed answer.

SOLUTION

Seven steps, and a person signs off

1. Ask

A resident's question comes in.

2. Screen

Personal data is redacted before storage. Requests to get around the rules are refused.

3. Search

1,024 passages from 45 official documents, plus open data through Zetaris.

4. Draft

NVIDIA Nemotron writes the answer. Every sentence points at a source.

5. Check

Quotes must match word for word, numbers must be in the evidence. Failing sentences are cut.

6. Approve

A person reads it with every quote highlighted on its page. Nothing goes out before.

7. Remember

The decision goes to Meterless H-MEM. A rejection changes the next similar answer.

WORKFLOW AND ARCHITECTURE

How it fits together

Question
From a resident, through the service team

→

Screen
Redact personal data, refuse rule-dodging

→

Recall
Past reviews of similar questions
Meterless H-MEM

→

Search
BM25 + Nemotron embeddings, plus open-data queries
NVIDIA
Zetaris

→

Draft
Every sentence cites a passage or a query
Nemotron 3 Nano

→

Check
Quotes word for word, numbers in evidence, queries rerun
Zetaris

→

Reviewer
Approve or reject, with quotes highlighted on the page

→

Answer sent
Only the sentences that passed

Zetaris Lightning database `dalil`

53 documents, 349 pages, 1,024 passages and four Bayanat open datasets. Search reads the passages; the checker reruns every cited query here.

Meterless H-MEM trust ledger

Every decision with the passages it cited. A rejection keeps those passages out of the next similar answer, and its note goes to the drafter.

Every step goes to an append-only audit trail. Everything runs on one server, models included.

TECHNICAL STACK

What it runs on

App	Python 3.13, FastAPI, SQLite audit trail, plain HTML, CSS and JavaScript screen
Zetaris	Lightning 2.4.3.1, the free local version: documents, pages, passages and four Bayanat open datasets in one database
Meterless	H-MEM reference engine on Node: review memory and trust ledger
NVIDIA	Nemotron 3 Nano 4B drafts, Nemotron-3-Embed-1B for dense search, both local through llama.cpp
Search	BM25 fused with the Nemotron embeddings
Size	45 documents, 349 pages, 1,024 passages, 4 open datasets, 237 tests

SPONSOR TECHNOLOGY

How Zetaris and Meterless are used

Zetaris

Documents, pages, passages and four Bayanat open datasets live in one Lightning database.

A number from open data is cited as a Zetaris query. The checker reruns that query before the sentence may stay.

Zetaris proves every passage equals its slice of the page: 1,024 of 1,024.

Meterless H-MEM

Every approval and rejection lands in H-MEM's append-only trust ledger, with the passages it cited.

Before drafting, Dalil recalls reviews of similar questions and leaves out passages a reviewer rejected.

WHAT IS NEW

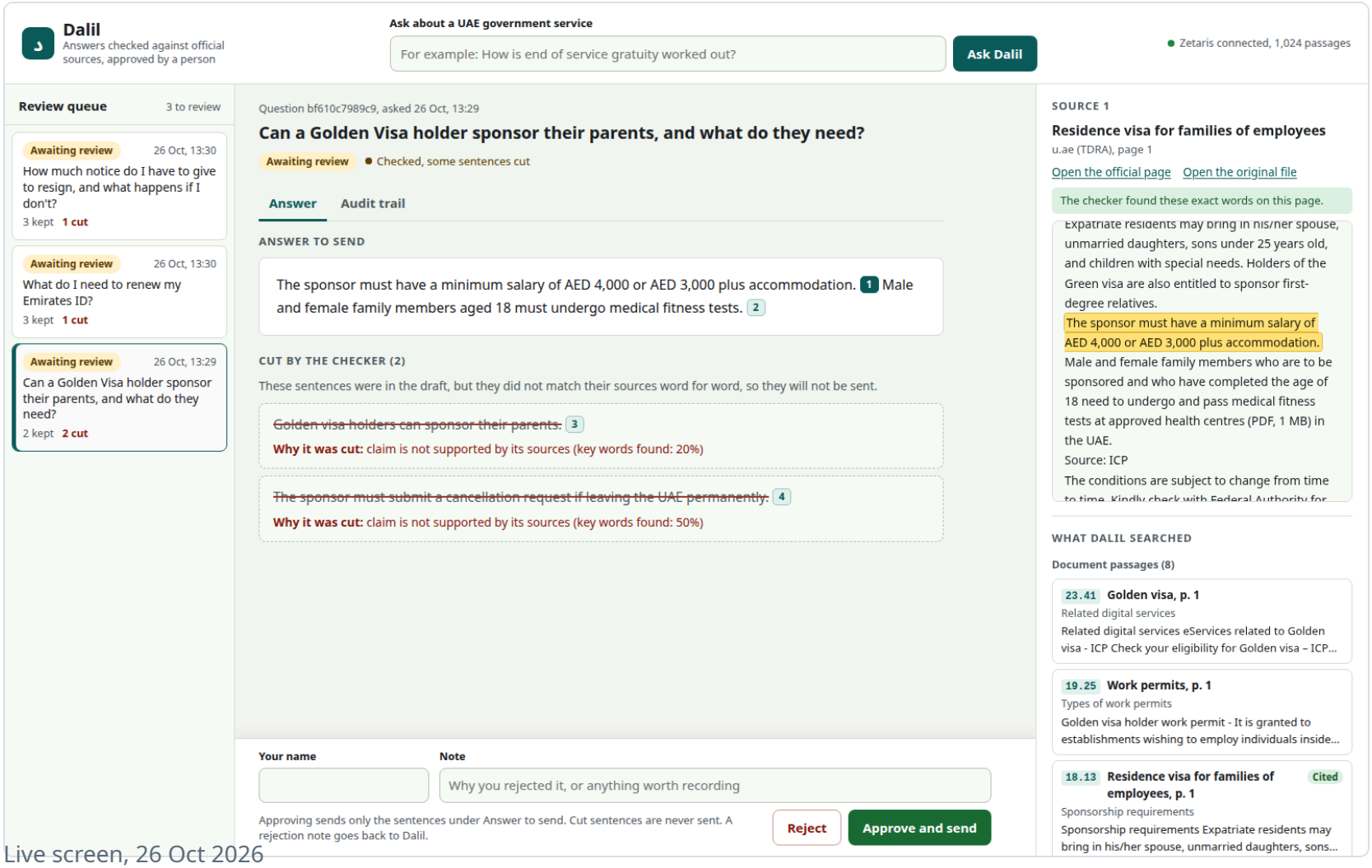
A citation you can rerun

- A quote maps to an exact character span on its page. A number maps to a Zetaris query.
- The checker trusts neither the model's quotes nor its SQL. Dalil writes each quote itself and refuses a number the evidence lacks.
- A reviewer's rejection changes the next answer to a similar question.

> Model draft: ~~"Golden visa holders can sponsor their parents."~~
> Cut by the checker: the cited source names spouses and children.

PRODUCT

The reviewer screen



- The answer to send, each sentence with a numbered citation.
- Below it, the sentences the checker cut and why. "Golden visa holders can sponsor their parents" never reaches the resident.
- On the right, the cited words highlighted on the official page.
- Approve or reject at the bottom. The audit trail tab replays every step.

Live screen, 26 Oct 2026

MEASURED, 26 OCT 2026

Results and limits

10 of 10

test questions cited an expected document. It was 9 of 10 in the 24 Oct run. All 10 passed the checker, 2 with a claim cut.

4 of 4

error scenarios behaved as intended: no invented fee, personal data redacted, outdated law flagged, rule-dodging refused

1,014

of 1,014 real passages: every reflowed quote found, every altered quote caught

- The team wrote the ten test questions, so the set is small.
- The checker tests quotes, numbers, sources and key words, not meaning. A person reads every answer.
- On CPU, the median answer takes 41 seconds. A GPU would be far faster.

NEXT

Where Dalil goes from here

- **ITU AI for Good, UAE:** the same agent on ITU's own knowledge-base code, with seven tested error scenarios. Due 10 Nov.
- **Arabic answers.**
- **A GPU,** to bring an answer well under the current time on CPU.
- **A company's own documents** in place of government ones, for business use.

Dalil

Answers about UAE government services, with every sentence tied to the official text it came from, and a person signing off before anything goes out.

github.com/sami-abdul/dalil

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